

IBM Data Analyst Capstone Project

Analysis on Emerging Technology Skills and Trends

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IBM Data Analyst Professional Certificate | 2026

OUTLINE

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EXECUTIVE SUMMARY

Data Sources: Stack Overflow Developer Survey 2019 (~11,500 respondents), Jobs API, and web-scraped salary data.

Languages: JavaScript dominates current usage. Python is rapidly rising — ranked #3 in desired languages for next year, up from #5 today.

Databases: MySQL leads current usage, but PostgreSQL is the #1 most desired database for next year — a clear shift underway.

Tools: Visual Studio Code is the overwhelming IDE of choice with nearly 6,500 users.

Compensation: Most developers earn \$0–\$75K. Doctoral degree holders earn the highest median salaries, though experience often outweighs credentials.

Demographics: The US has the largest share of respondents. ~97% are employed full-time. 60% prefer working from an office.

INTRODUCTION

PURPOSE

This project uses data analytics to identify current and emerging technology skill trends across the global IT industry. By analysing real developer survey data alongside job postings and salary information, the goal is to produce insights that help organisations stay competitive and help professionals prioritise the right skills.

KEY QUESTIONS

1. Which programming languages are most in demand today?
2. What are the most in-demand database skills?
3. Which IDEs are developers using and planning to adopt?
4. How do salary, age, and education relate?

TARGET AUDIENCE

- IT Professionals & Developers
- HR Managers & Recruitment Teams
- Training & Development Departments
- Business Leaders in Technology

METHODOLOGY

1



Web Scraping

Scraped programming language salary data from IBM-hosted HTML pages using Python BeautifulSoup.

2



API Collection

Called the IBM Jobs API to count job postings by technology skill and city location.

3



Data Wrangling

Removed 154 duplicate rows. Imputed missing values with mode (WorkLoc) and median (salary). Final dataset: ~11,400 rows.

4



EDA

Analysed distributions, outliers (IQR method), and correlations across salary, age, and education.

5



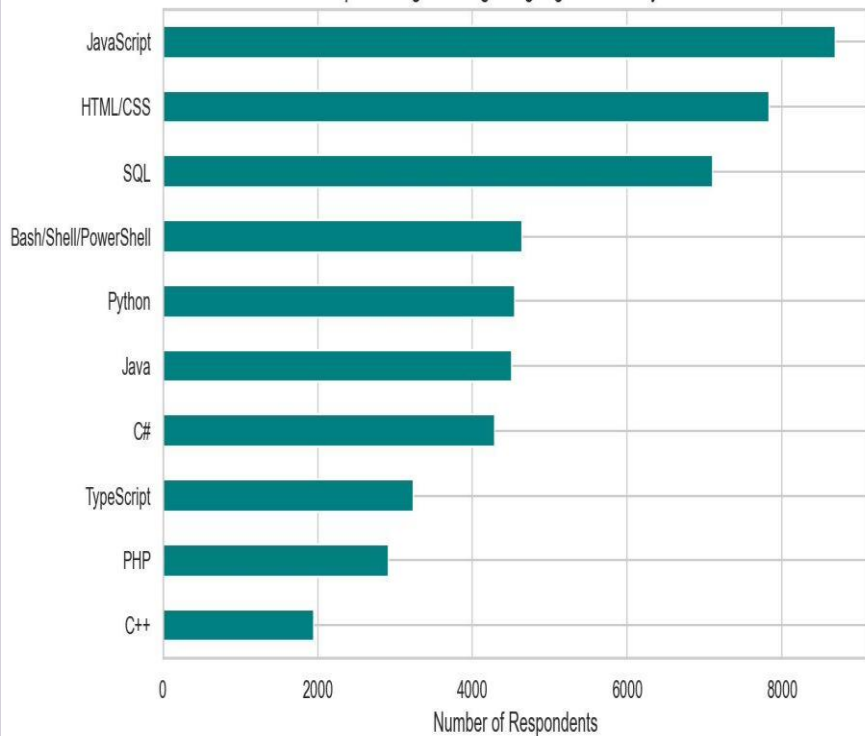
Visualisation

Built 11 charts in Python (Matplotlib/Seaborn) and 3 interactive dashboard tabs in IBM Cognos Analytics.

RESULTS — PROGRAMMING LANGUAGE TRENDS

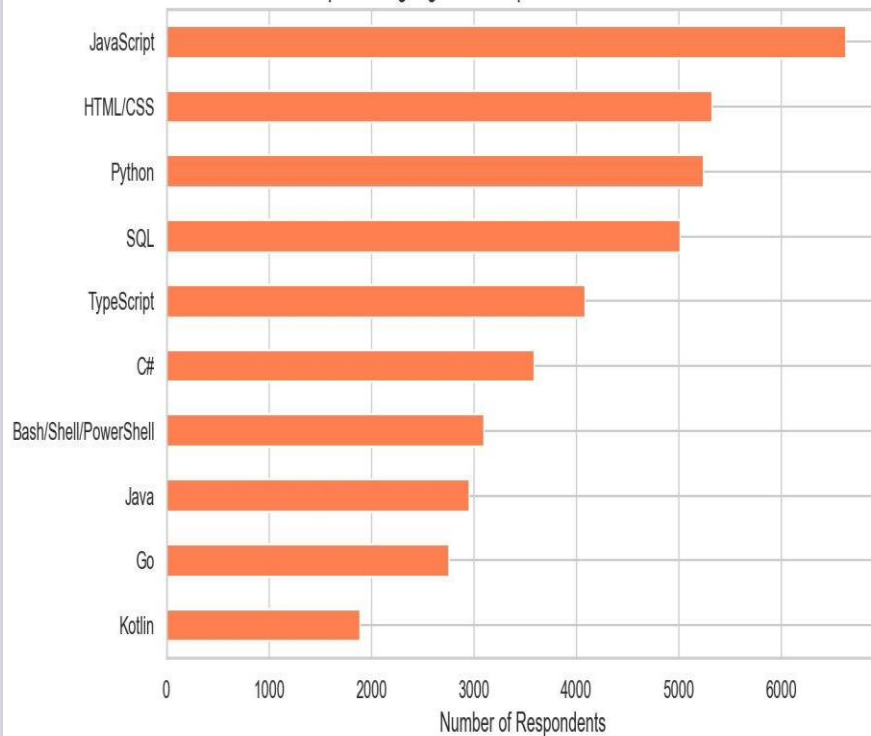
Current Year

Top 10 Programming Languages Currently Used



Next Year (Desired)

Top 10 Languages Developers Want to Use Next Year



PROGRAMMING LANGUAGE TRENDS — FINDINGS & IMPLICATIONS

FINDINGS

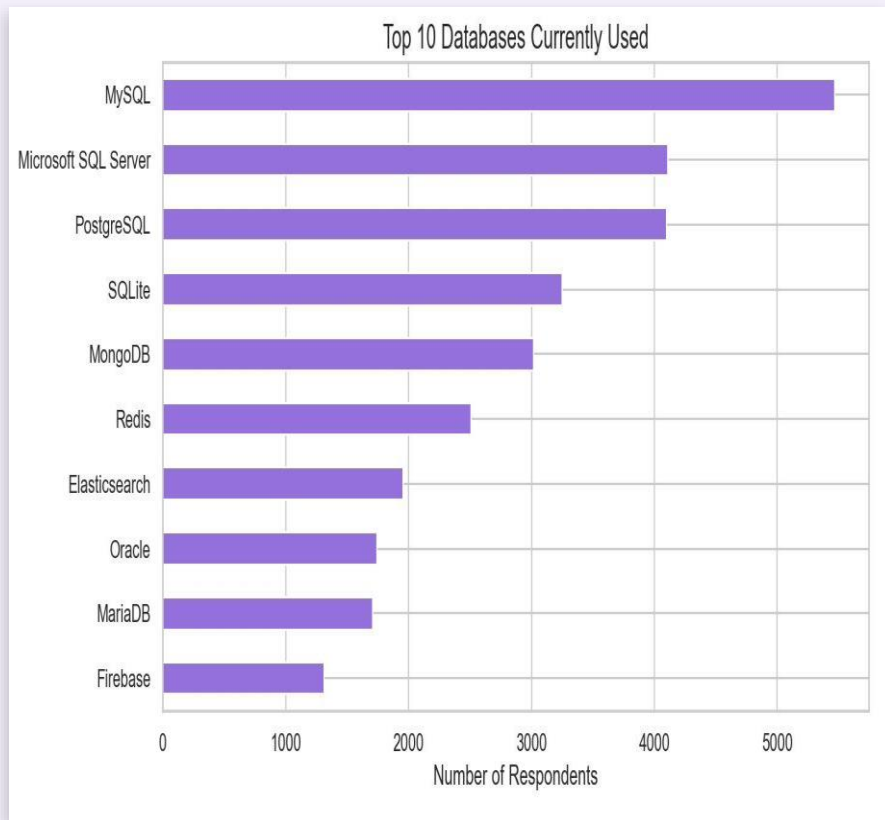
- JavaScript is the most-used language (9,000+ respondents) and retains #1 status in desired usage.
- HTML/CSS and SQL remain top-3 in both current and desired lists — essential for full-stack and data work.
- Python jumps from #5 currently to #3 desired, reflecting strong demand in AI, data science, and automation.
- TypeScript and Go are new entrants in desired usage — indicating a shift toward type-safe and scalable modern languages.

IMPLICATIONS

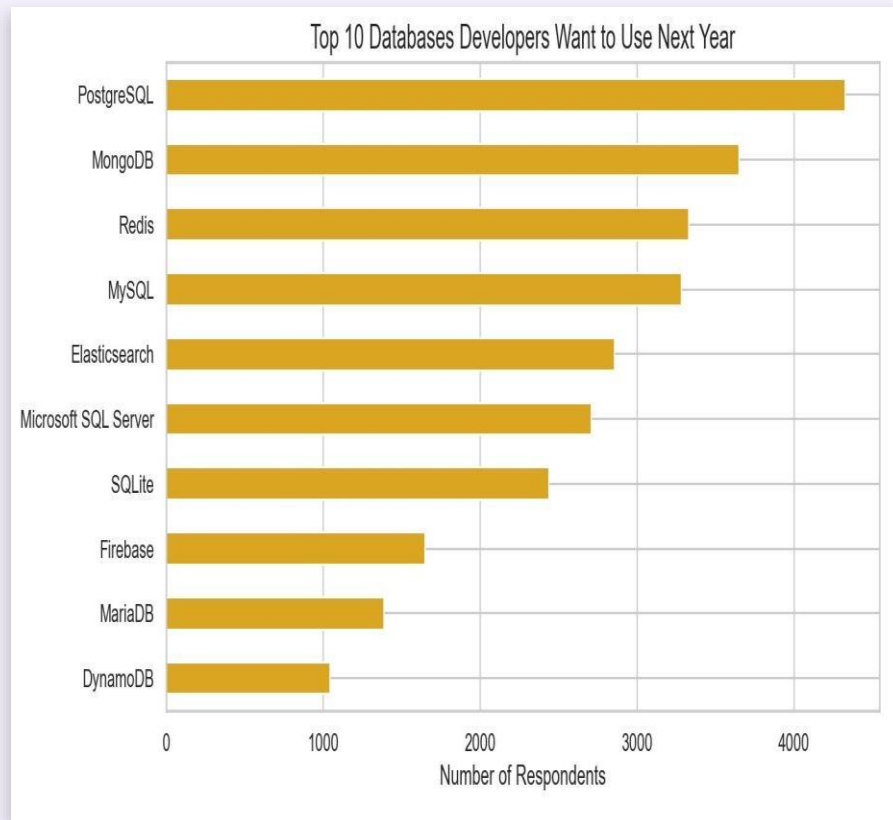
- Train teams in Python now — its growth trajectory in AI/ML will make it non-optional within 2–3 years.
- Web development skills (JavaScript, HTML/CSS) remain the highest-demand category globally.
- SQL is still critical for data professionals — it appears in both current and desired top-5 lists.
- Invest in TypeScript training — as JavaScript codebases scale, TypeScript adoption will accelerate rapidly.

RESULTS — DATABASE TRENDS

Current Year



Next Year (Desired)



DATABASE TRENDS — FINDINGS & IMPLICATIONS

FINDINGS

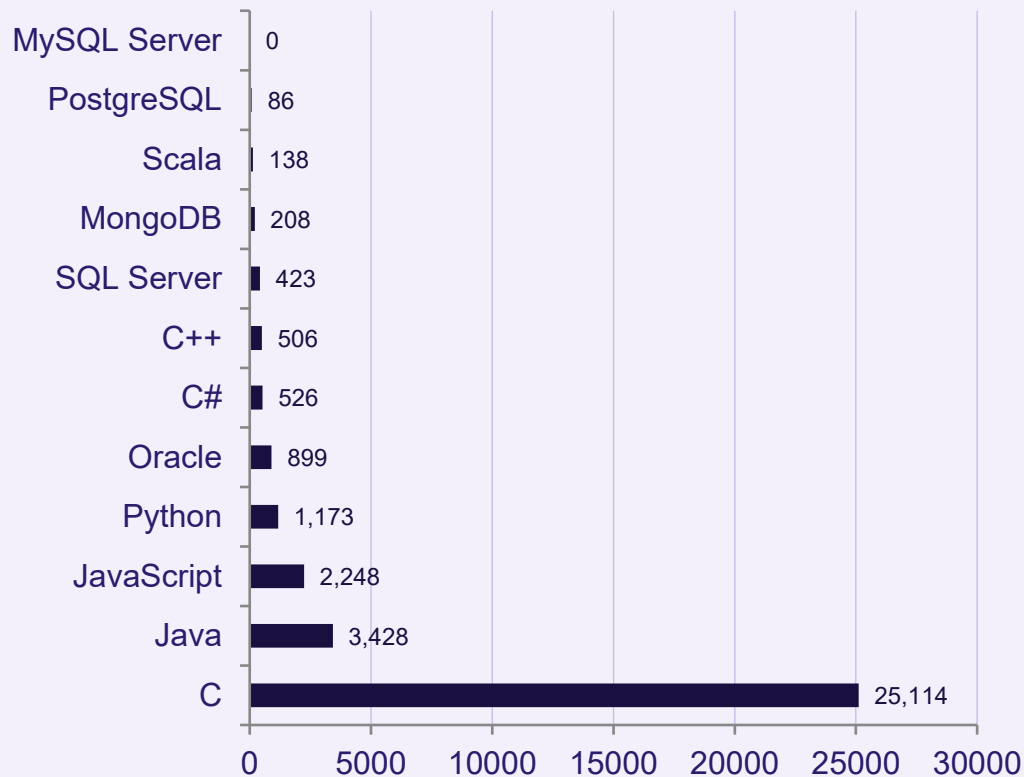
- MySQL is the most widely used database today (5,500+ respondents), ahead of MS SQL Server and PostgreSQL.
- PostgreSQL overtakes MySQL as the #1 desired database for next year — a significant leadership change.
- MongoDB and Redis are rising fast in desired usage, confirming growing NoSQL and caching adoption.
- Oracle did not appear in the top 5 for either current or desired use — it is losing relevance in the developer community.

IMPLICATIONS

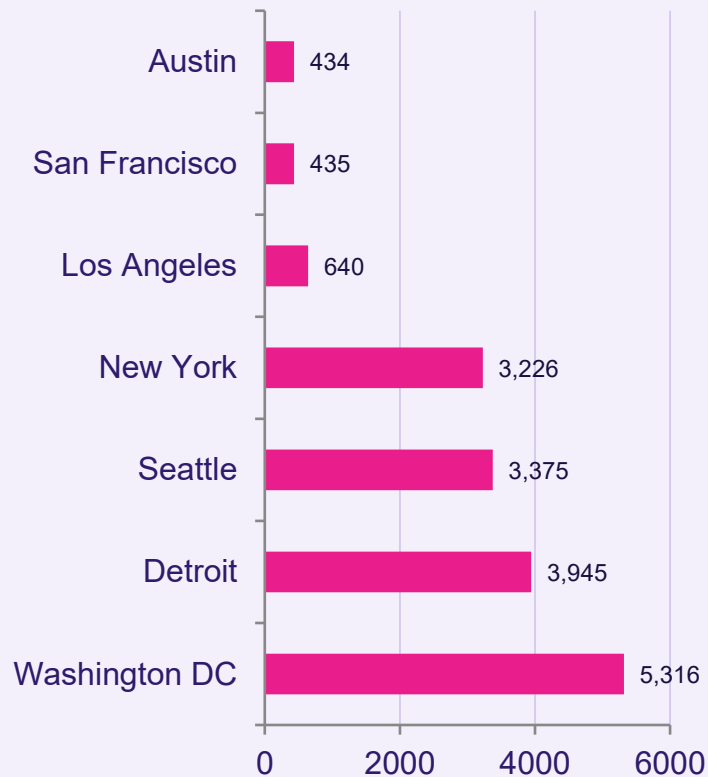
- Teams should begin upskilling in PostgreSQL — it is positioned to replace MySQL as the default relational database.
- MongoDB and Redis skills will be critical for modern cloud-native and microservice architectures.
- Open-source databases dominate — companies should align infrastructure decisions with open-source tools.
- Oracle SQL training budgets can be gradually redirected to PostgreSQL and cloud database platforms.

JOB POSTINGS — BY TECHNOLOGY & LOCATION

Job Postings by Technology Skill

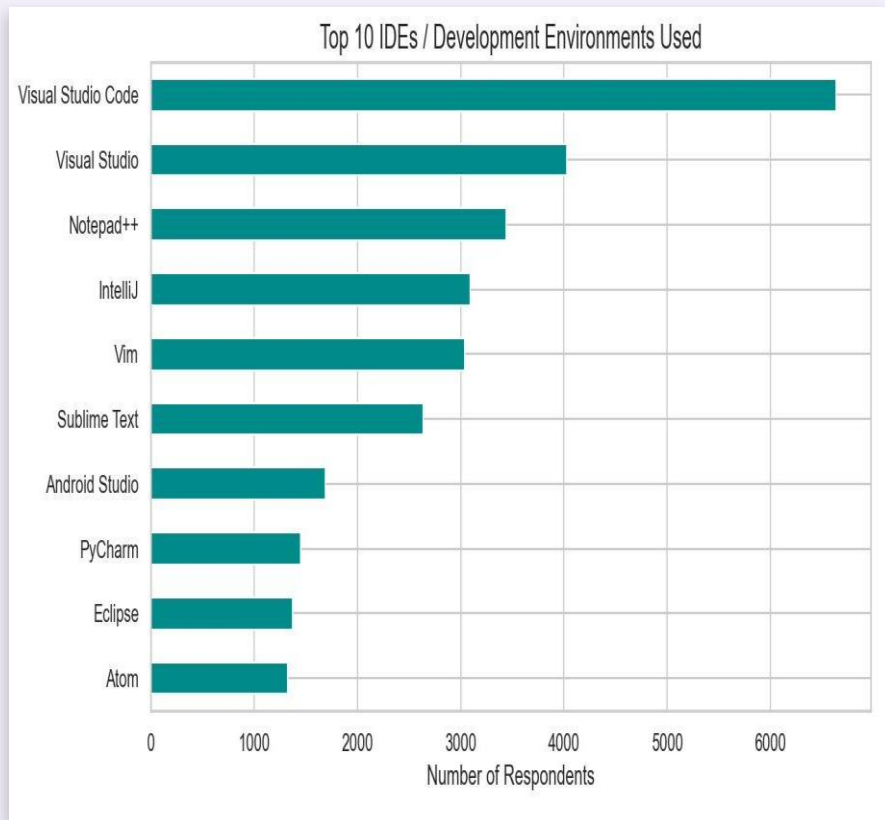


Job Postings by City

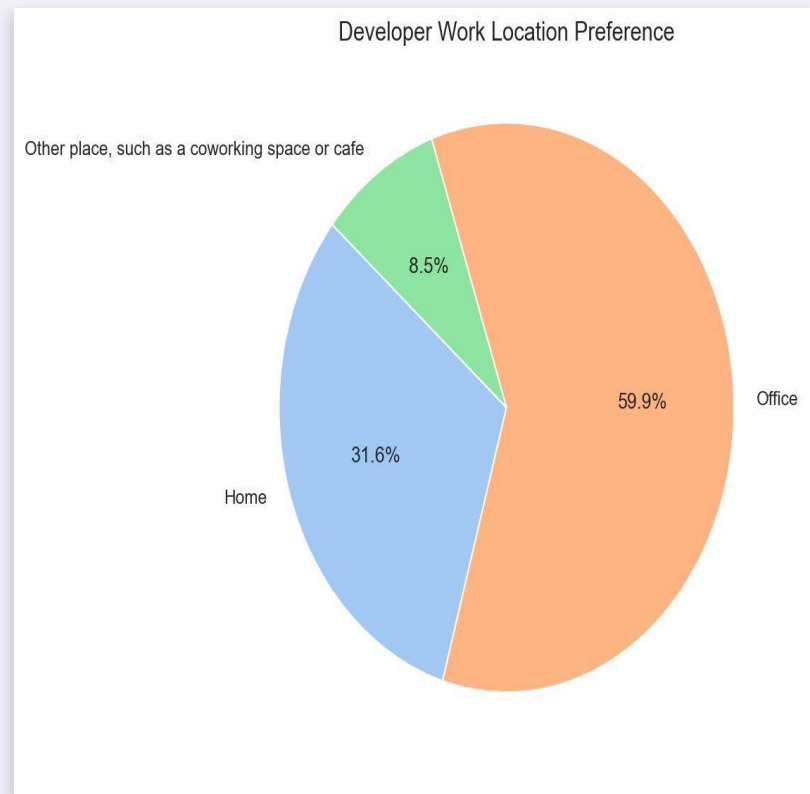


ADDITIONAL TRENDS — IDEs & WORK LOCATION

Top 10 IDEs Used



Work Location Preference

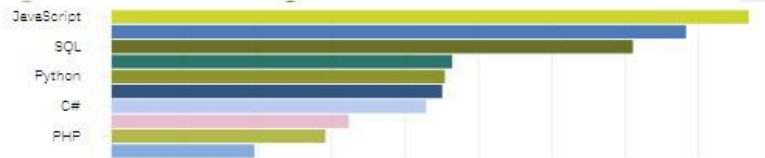


DASHBOARD — TAB 1: CURRENT TECHNOLOGY USAGE

Top 10 LanguageWorkedWith

LanguageWorkedWith

- Bash/Shell/PowerShell
- C++
- Java
- C#
- HTML/CSS
- JavaScript



Top 10 DatabaseWorkedWith

DatabaseWorkedWith

- Elasticsearch
- MariaDB
- MongoDB
- Oracle
- Firebase
- Microsoft SQL Server
- MySQL
- PostgreSQL



PlatformWorkedWith

PlatformWorkedWith

PlatformWorkedWith



Top 10 WebframeWorkedWith

WebFrameWorke...

WebFrameWorkedWith

- ASP.NET
- Django
- Angular/Angular.js
- Express

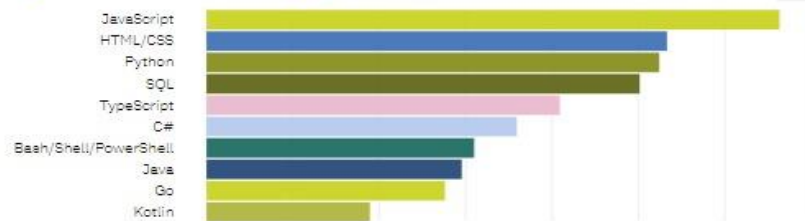


DASHBOARD — TAB 2: FUTURE TECHNOLOGY TRENDS

Top 10 LanguageDesireNextYear

LanguageDesireNextYear

● Bash/Shell/PowerShell ● C#
● Go ● HTML/CSS



Top 10 Most Desired Databases

DatabaseDesireNextYear

● DynamoDB ● Elasticsearch
● Firebase ● MariaDB



Most desired Platforms next year

PlatformDesireN...



PlatformDesireN...



Most Desired Web frames Next Year

WebFrameDesire...



WebFrameDesireNextYear

● (no value)
● ASP.NET



DASHBOARD — TAB 3: DEMOGRAPHICS

Percentage of Respondents by gender

Gender

Woman Man

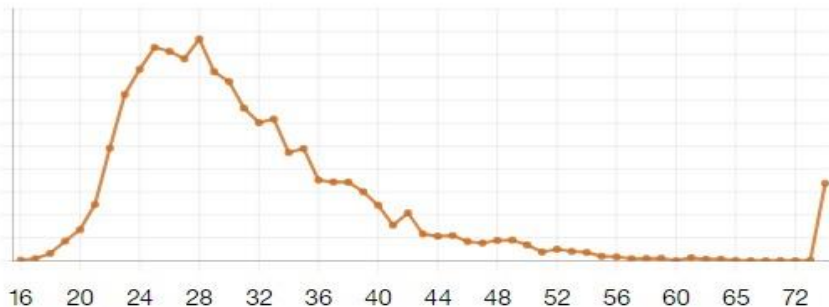


Respondents by Country

Respondent (Sum)



Respondents by Age

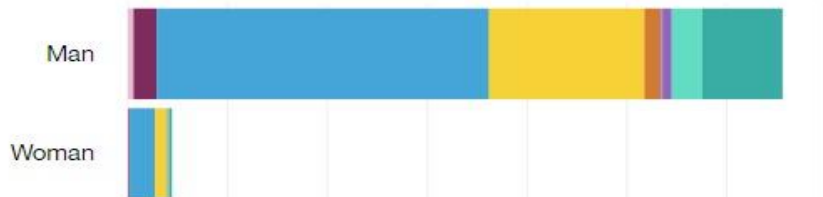


Respondents by Formal Education Level

Education level

(no value)

Associate degree



DASHBOARD INSIGHTS



Tab 1 — Current Technology

JavaScript, HTML/CSS, SQL dominate current usage across all developer types. • VS Code leads IDE adoption by a wide margin over all other tools. • MySQL is the #1 database in active production use globally.



Tab 2 — Future Trends

Python overtakes SQL in desired language rankings for the coming year. • PostgreSQL is the most wanted database — signalling a shift from MySQL. • TypeScript and Go are emerging as the next-generation language choices.



Tab 3 — Demographics

The US contributes the most respondents; India and UK follow. • Most developers are aged 20–35, with salary growth plateauing around age 40. • ~97% are full-time employed; Bachelor's degree is the most common qualification.

OVERALL FINDINGS & IMPLICATIONS

11,500+

Survey
Respondents

#1

JavaScript
Most Used Language

PostgreSQL

Most Desired
Database 2024

VS Code

Dominant
IDE Choice

KEY FINDINGS

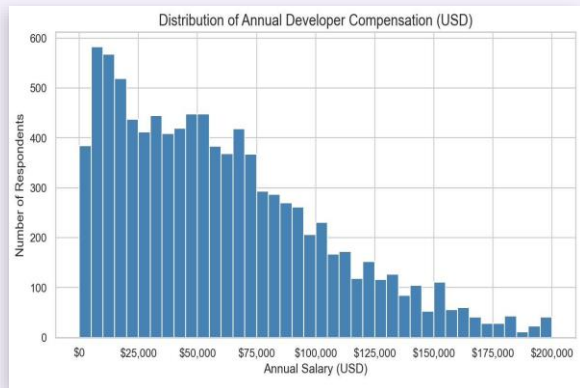
- Web languages (JavaScript, HTML/CSS) have the highest global demand.
- Python is growing faster than any other language year-over-year.
- PostgreSQL is overtaking MySQL as the relational database of choice.
- Most developers are aged 20–35 and hold a Bachelor's degree.
- The US, India, and UK are the primary hubs for IT talent.

BROADER IMPLICATIONS

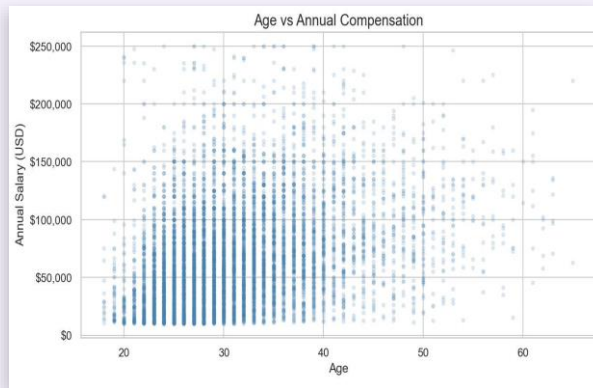
- Tech training programs must prioritise Python and TypeScript to stay relevant.
- NoSQL skills (MongoDB, Redis) are no longer optional for backend developers.
- Remote and hybrid work policies are needed to attract top talent globally.
- Skills-based hiring should complement or replace degree-only requirements.
- Growing developer communities in Asia and South America present new talent pools.

APPENDIX — ADDITIONAL VISUALIZATIONS

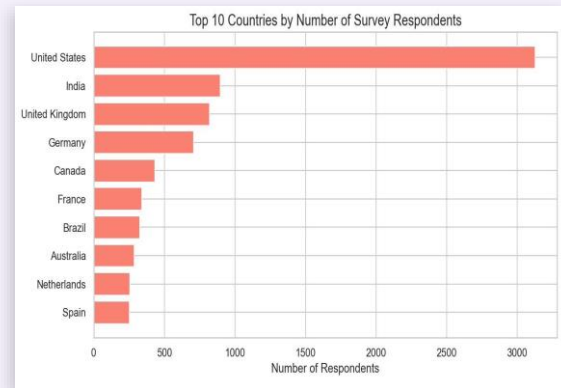
Salary Histogram



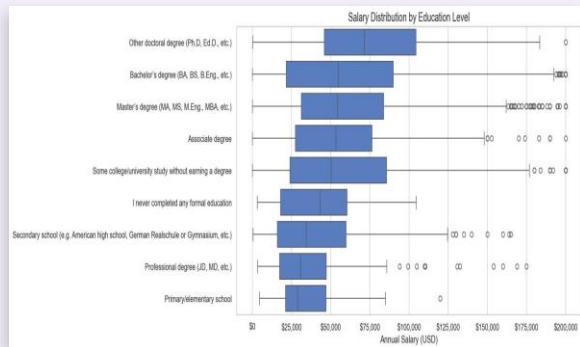
Age vs Salary Scatter



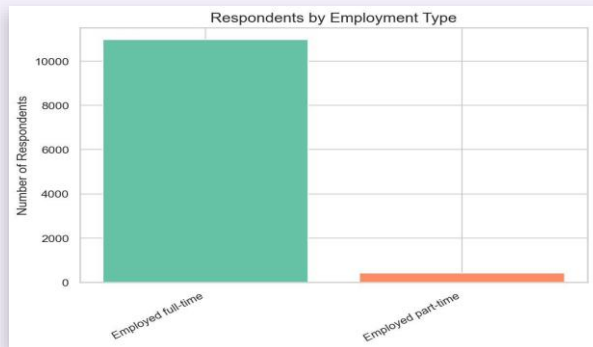
Top 10 Countries



Salary by Education Level



Employment Type



Dataset Stats

Respondents: ~11,552
Countries: 183
Columns: 64
Duplicates removed: 154
Survey year: 2019

CONCLUSION



Technology trends move fast — continuous upskilling is no longer optional, it is the baseline expectation for IT professionals.



Python and PostgreSQL are the two most important skills to invest in heading into the next 1–2 years.



JavaScript and SQL remain evergreen — foundational skills that continue to anchor the entire technology ecosystem.



Visual Studio Code has become the universal standard development environment across languages and disciplines.



Data-driven decisions about technology investment lead to better hiring, training, and retention outcomes.

Thank You

Thank You